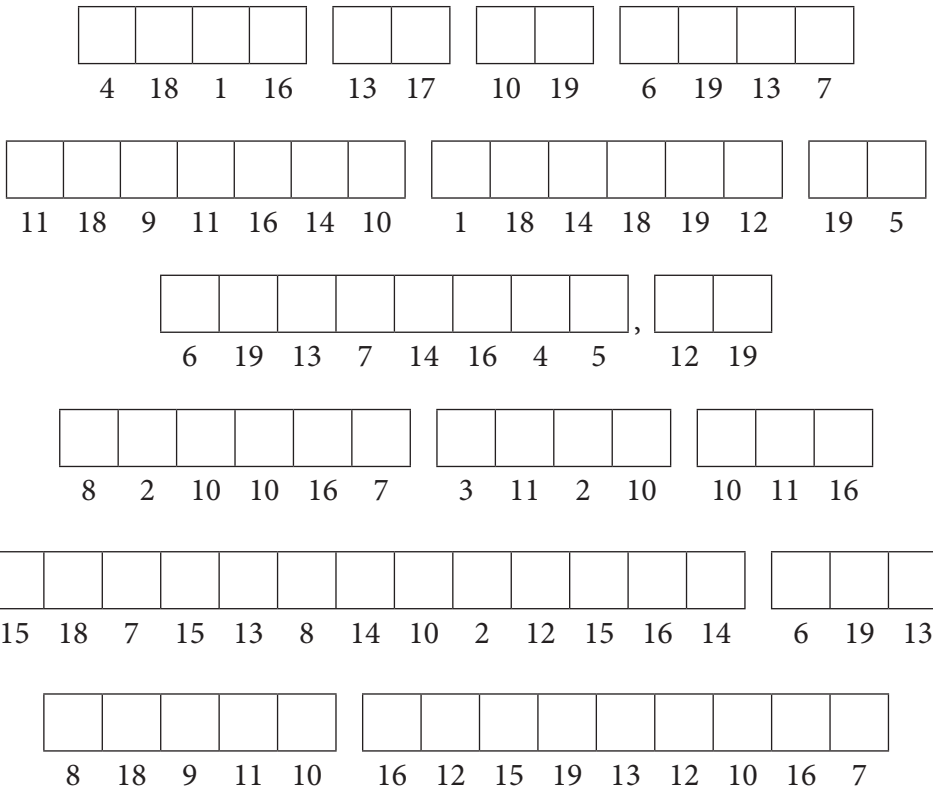
Name: _____

WORKSHEET

Arithmetic series



Ayn Rand

Rule

A finite arithmetic progression is called an **arithmetic series**.
The sum of the first n terms in an arithmetic progression can be calculated in two ways:

$$S_n = \frac{n}{2}[2a + (n-1)d] \text{ or } S_n = \frac{n}{2}[a + l]$$

where a is the first term
l is the last term
d is the common difference between terms.

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Questions

Find the sum of each of the following arithmetic series.

1 The first 5 terms of 6, 10, 14, …

2 The first 30 terms of 17, 18, 19, …

3 The first 24 terms of 100, 95, 90, …

4 The first 51 terms of $\frac{1}{2}$, 1, $1\frac{1}{2}$, …

5 The first 27 terms of 43, 38, 33, …

6 $14 + 11 + 8 + \ldots - 1 - 6$

7 $34 + 45 + 56 + \ldots + 276$

8 $85 + 78 + 71 + \ldots - 559$

9 $-5 + -2\frac{1}{3} + \frac{1}{3} + \ldots + 83$

10 All positive even numbers less than 200

11 All numbers between 148 and 302 that are divisible by 5

12 The first $3n - 2$ terms of 1, 7, 13, …

13 The first $n + 1$ terms of 3, 5, 7, …

14 The first $2n - 5$ terms of 10, 7, 4, …

15 How many terms of the series $5 + 8 + 11 + \ldots$ need to be added together to equal 549?

16 How many terms of the series $120 + 114 + 108 + \ldots$ need to be added together to equal 990?

17 If the 1st term of an arithmetic sequence is 43 and the 12th term is -12, find the sum of the first 20 terms.

18 If the 3rd term of an arithmetic sequence is 1941 and the 22nd term is 1371, find the sum of the first 14 terms.

19 If the 7th term of an arithmetic sequence is 15 and the 11th term is 23, find the sum of the first 50 terms.

Solutions

A 945

C 18

E 11 or 30

F -594

G 1326

H 6975

I 25 284

L 663

M $-22\,041$

N $27n^2 - 42n + 16$

O 2600

P -90

R 3565

S $-6n^2 + 53n - 95$

T 9900

U $n^2 + 4n + 3$

V 70

W 1020

Y -11

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